

END-SEM EXAM NOTES

Violence Detection Using Affective / Upper-Body Skeleton Features

Based primarily on the uploaded research paper; extra exam formulas are clearly marked as general ML knowledge.

1. Paper at a Glance — Memorize This First

Item	Exam-ready answer
Research problem	Detect violence/abusive behaviour from video using body/arm movement and affect-related behavioural cues.
Main idea	Extract 2D human skeleton keypoints with OpenPose, derive upper-body/arm features, then classify violence vs non-violence.
Dataset	RWF2000: 2,000 videos; 1,000 violent + 1,000 non-violent; 80% train / 20% test; each video has 150 frames.
Input modality	RGB videos → OpenPose → 2D skeletal keypoints.
Features	Arm joint locations, elbow/shoulder angles, temporal patches of elbow/wrist motion.
Main classifier	Fine KNN, evaluated with 5-fold cross-validation during training.
Best training result	Joint-location features + Fine KNN: 98.6% overall training accuracy.
Test result	Joint-location + Fine KNN: 71.2%; angles: 65.5%; temporal patches: 67.0%.
Benchmark	Flow Gated Network / deep learning: 85.75% on RWF2000.
Main limitation	OpenPose may fail on occluded/low-resolution scenes; some non-violent videos contain no people, so skeleton features are missing.
Future work	Add Quantity of Motion (QoM), Motion History Image (MHI), Optical Flow, and test other datasets.

2. The Six Questions You Should Prepare for Every Research Paper

1) Research Question

What problem is the paper trying to solve? Here: can violence/abusive behaviour be detected using low-level upper-body behavioural cues extracted from 2D skeletons, rather than relying only on large visible fights or facial expressions?

2) Significance

Violence systems often emphasize vigorous two-person fights. Abuse can be subtler. Adding emotion/affect-related body cues may help surveillance systems recognize aggressive behaviour.

3) Indian-context example

Example: CCTV/video monitoring in an Indian railway station, metro station, college campus, hospital, or public gathering could use the method as an additional alert mechanism for suspected physical aggression. It should be treated as decision support, not automatic proof of violence.

4) Dataset

RWF2000 has 2,000 YouTube-derived real-life violence/non-violence videos. The paper reports 1,000 in each class, 80/20 train-test split, and 150 frames per video. The paper does not report a number of human subjects; therefore do not invent one.

5) Methodology

OpenPose extracts 25 2D keypoints. Three feature families are computed: joint locations, joint angles, and temporal patches. Fine KNN classifiers are trained and compared using 5-fold cross-validation.

6) Results

Training results were very high, but held-out test performance dropped substantially. Best reported test accuracy among the proposed feature sets was 71.2%, versus 85.75% for the cited deep-learning Flow Gated Network benchmark.

3. Research Question — How to Explain in an Exam

Background: Traditional violence detection often uses motion, trajectory, appearance, or handcrafted video descriptors. Many systems focus on obvious fights. The paper argues that affective/anger-related upper-body behaviour can provide useful cues.

Core question: Can low-level features derived from upper-body 2D skeletal keypoints capture subtle arm movements associated with aggressive behaviour and classify violent versus non-violent videos?

Hypothesis/intuitive idea: Anger/aggression can be reflected by arm bending, arm extension, shoulder elevation, and movement over time. These behaviours can be represented numerically from skeleton coordinates.

4. Significance

- Improves the conceptual scope of surveillance from only large fights to smaller behavioural cues.
- Uses body information rather than depending mainly on facial expressions.
- Skeleton features can reduce dependence on raw RGB appearance and focus on geometry/motion.
- Fine KNN produced high training accuracy with minimal training time in the reported experiments.
- However, real-world robustness remains a major issue because surveillance video may be low-resolution or occluded.

5. Dataset — RWF2000

Property	Value
Dataset	RWF2000
Total videos	2,000
Violent videos	1,000
Non-violent videos	1,000
Train/test split	80% / 20%
Approx. videos if split is applied directly	1,600 train / 400 test
Frames per video	150
Source	YouTube; real-life violence scenes
Video characteristics	Surveillance-camera style; low resolution and occlusion can occur
Modality	RGB video; 2D skeleton extracted from RGB using OpenPose
Number of human subjects	Not reported in the paper

6. OpenPose and 2D Skeleton

OpenPose is used to extract 2D skeletal joint locations from RGB video. The paper refers to 25 keypoints. For the proposed arm features, four keypoints are important: two from the right arm and two from the left arm, particularly elbow and wrist coordinates. Shoulder and mid-hip points are also used for angle computation.

Why skeleton? It represents body geometry and movement while reducing the problem to numerical coordinates. This is useful when the research question is about body/arm behaviour.

7. Feature Engineering — Very Important

A) Joint-location features

Use the x-y coordinates of elbow and wrist for both arms. These describe the frame-by-frame position of the arms.

B) Joint-angle features

Angles are calculated to represent arm bending/extension and shoulder movement. The paper uses the cosine rule.

Elbow angle:

$$\theta_{\text{elbow}} = \cos^{-1}[(s_e^2 + e_w^2 - s_w^2) / (2 s_e e_w)]$$

where s_e = shoulder–elbow distance, e_w = elbow–wrist distance, s_w = shoulder–wrist distance.

Shoulder angle:

$$\theta_{\text{shoulder}} = \cos^{-1}[(h_s^2 + s_e^2 - h_e^2) / (2 h_s s_e)]$$

where h_s = mid-hip–shoulder distance, s_e = shoulder–elbow distance, h_e = mid-hip–elbow distance.

C) Temporal-patch features

The paper uses a 20-frame temporal window. It measures how far a keypoint moves from one frame to the next.

For elbow:

$$d_t(\text{elbow}) = \sqrt{(e_{i,t} - e_{i,t-1})^2 + (e_{j,t} - e_{j,t-1})^2}$$

For wrist:

$$d_t(\text{wrist}) = \sqrt{(w_{i,t} - w_{i,t-1})^2 + (w_{j,t} - w_{j,t-1})^2}$$

Temporal patch for elbow:

$$TP_e = \sum_{\text{from } t-20 \text{ to } t} d_t(\text{elbow})$$

Temporal patch for wrist:

$$TP_w = \sum_{\text{from } t-20 \text{ to } t} d_t(\text{wrist})$$

Interpretation: a larger temporal-patch value means greater accumulated movement of that joint during the window.

8. Why Fine KNN?

The paper trains multiple classifiers using the extracted feature sets and reports that KNN networks performed very well for the frame-by-frame features. Fine KNN was used for the reported feature-set comparison. It is suitable when similar feature vectors tend to belong to the same class and it is simple/fast to train because much of the work happens during prediction.

Alternative model: SVM. The related-work section specifically notes that SVM is commonly used for violence detection with handcrafted features such as Violent Flows, OHOF, STIP/HOG/HOF, etc.

Other alternatives from the paper: K-means (unsupervised), 3D CNN/C3D, ConvLSTM, ResNet50 + LSTM, Flow Gated Network, and hybrid multi-stream CNN approaches.

9. Machine Learning Concepts You Can Be Asked

Supervised learning: Training data has labels. The model learns a mapping from features to classes. Violence detection is commonly treated as supervised classification.

Unsupervised learning: Training data has no class labels. K-means is mentioned as an example for grouping violence-related features.

Handcrafted features: Features are designed using domain knowledge, e.g., motion, trajectory, appearance, joint angles, and temporal movement.

Deep learning: CNN/LSTM-based systems learn representations automatically from data. Examples in the paper include 3D CNN, ConvLSTM, C3D, and ResNet50 + LSTM.

Hybrid method: Combines deep-learning modules with handcrafted features and/or traditional ML classifiers. The proposed work is hybrid because OpenPose/deep architecture extracts skeletons, handcrafted behavioural features are computed, and ML classification is then performed.

10. Experiments

Pipeline:

RGB video → OpenPose → 2D skeleton → feature extraction → Fine KNN → violence/non-violence classification

Implementation reported: OpenPose extraction in Python using an NVIDIA RTX2060 GPU; feature extraction in MATLAB; MATLAB Classification Learners app for classification; 5-fold cross-validation for training comparisons.

11. Results — Memorize the Numbers

Feature set	Violence accuracy	Non-violence accuracy	Overall training accuracy	Training time
Joint location for arms	99.4%	96.4%	98.6%	5.8625 s
Joint angles	97.5%	82.2%	93.5%	12.744 s
Temporal patches	99.0%	91.3%	97.0%	5.4109 s

Method	Test accuracy
Our joint-location + Fine KNN	71.2%
Our joint-angle + Fine KNN	65.5%
Our temporal-patch + Fine KNN	67.0%
Cited Flow Gated Network	85.75%

12. How to Explain the Train–Test Drop

Training performance is high, but testing performance is much lower. The paper attributes this partly to non-violent videos that contain no people (for example, cars or other scenes). If no person is detected, fewer than half of the frames may contain usable 2D skeletons, so the proposed skeleton-based features become unavailable or weak.

Exam conclusion: High training accuracy does not guarantee generalization. Always discuss the held-out test result and the failure cases.

13. Accuracy and Other Important Exam Formulas

The following formulas are general classification formulas useful for interpreting results; they are not all explicitly given in the uploaded paper.

Accuracy: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$

Precision: $\text{Precision} = TP / (TP + FP)$

Recall / Sensitivity: $\text{Recall} = TP / (TP + FN)$

Specificity: $\text{Specificity} = TN / (TN + FP)$

F1-score: $F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Error rate: $\text{Error rate} = 1 - \text{Accuracy}$

KNN Euclidean distance: $d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}$

Mean accuracy in k-fold CV: $\text{CV accuracy} = (1/k) \sum_j \text{Accuracy}_j$

Confusion-matrix meanings

TP = violent correctly predicted as violent; TN = non-violent correctly predicted as non-violent; FP = non-violent incorrectly predicted as violent; FN = violent incorrectly predicted as non-violent.

14. Related-Work Models — Know the Name + One-Line Purpose

Feature/model	What it captures / why used
MEI	Motion Energy Image; summarizes where motion occurs across frames.
MHI	Motion History Image; represents motion history over time.
ViF	Violent Flows; uses changes in optical-flow vector magnitudes over time.
OHOF	Oriented Histogram of Optical Flow; uses motion orientation information.
STIP	Space-Time Interest Points; captures local spatial-temporal changes.
HOG/HOF	Histogram of Oriented Gradients / Histogram of Optical Flow; appearance/shape and motion descriptors.
MoSIFT	Motion-oriented extension of SIFT used with motion/appearance information.
SVM	Supervised classifier commonly paired with handcrafted violence features.
K-means	Unsupervised clustering method used in some violence-detection work.
3D CNN / C3D	Learns spatial + temporal video representations using 3D convolutions.
ConvLSTM	Combines convolution with recurrent temporal modelling.
ResNet50 + LSTM	CNN extracts visual representations; LSTM models temporal sequence.
Flow Gated Network	Deep-learning benchmark reported at 85.75% on RWF2000.

15. Indian Context — Ready-Made Exam Example

Suppose CCTV cameras in an Indian railway/metro station observe a crowded platform. A system could extract body keypoints and monitor arm movement to flag a possible aggressive interaction. The alert could then be reviewed by security personnel. The method is especially relevant where faces are occluded or low-resolution, but the paper's own limitations mean a real deployment should combine skeleton features with motion/RGB cues and human verification.

16. Limitations — Very Important for Long Answers

- 2D skeleton extraction can be inaccurate under occlusion and low resolution.
- Some non-violent videos contain no people, so OpenPose cannot provide useful skeleton features.
- Testing accuracy is substantially below the cited benchmark.
- Skeleton-only features may miss scene/context information available in RGB and optical flow.
- The paper proposes adding QoM, MHI and Optical Flow and testing more datasets in future.

17. 5-Mark / 10-Mark Answer Structure

For 5 marks: Problem → dataset → features → model → best result → limitation.

For 10 marks: Research question → significance → Indian application → dataset/modality → OpenPose → three feature families + formulas → Fine KNN + alternative SVM/deep model → experimental setup → training vs testing results → limitation → future work.

18. Last-Minute Memorization Sheet

- ★ RWF2000 = 2000 videos = 1000 violent + 1000 non-violent.
- ★ 80/20 split → approximately 1600 train + 400 test if applied directly.
- ★ 150 frames/video.
- ★ OpenPose → 25 2D keypoints.
- ★ Main proposed features = joint locations + joint angles + temporal patches.
- ★ Temporal window = 20 frames.
- ★ Classifier = Fine KNN.

- ★ Cross-validation = 5-fold.
- ★ Best training overall = 98.6% (joint locations).
- ★ Best proposed test = 71.2% (joint locations).
- ★ Benchmark = Flow Gated Network, 85.75%.
- ★ Main failure reason = no person / poor skeleton extraction in some non-violent videos.
- ★ Future = QoM + MHI + Optical Flow + more datasets.

Source note: This study guide is grounded in the uploaded paper text. Claims about the paper's dataset, features, methodology, results, limitations, and future work are taken from that source. General ML formulas are included separately as exam-support material.